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Transistor gate structure and method of forming

Abstract

A device includes a first nanostructure; a second nanostructure over the first nanostructure; a high-k gate dielectric around the first nanostructure and the second nanostructure, the high-k gate dielectric having a first portion on a top surface of the first nanostructure and a second portion on a bottom surface of the second nanostructure; and a gate electrode over the high-k gate dielectric. The gate electrode comprises: a first work function metal around the first nanostructure and the second nanostructure, the first work function metal filling a region between the first portion of the high-k gate dielectric and the second portion of the high-k gate dielectric; and a tungsten layer over the first work function metal, the tungsten layer being free of fluorine.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE (1) This application is a continuation of U.S. application Ser. No. 17/854,175, filed on Jun. 30, 2022, which is a continuation of U.S. application Ser. No. 17/078,911, filed Oct. 23, 2020, now U.S. Pat. No. 11,437,240, issued on Sep. 6, 2022, which claims the benefit of U.S. Provisional Application No. 63/061,361, filed on Aug. 5, 2020, which application is hereby incorporated herein by reference.

BACKGROUND

(1) Semiconductor devices are used in a variety of electronic applications, such as, for example, personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductor layers of material over a semiconductor substrate, and patterning the various material layers using lithography to form circuit components and elements thereon.

(2) The semiconductor industry continues to improve the integration density of various electronic components (e.g., transistors, diodes, resistors, capacitors, etc.) by continual reductions in minimum feature size, which allow more components to be integrated into a given area. However, as the minimum features sizes are reduced, additional problems arise that should be addressed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates an example of a nanostructure field-effect transistor (nano-FET) in a three-dimensional view, in accordance with some embodiments.

(3) FIGS. 2, 3, 4, 5, 6A, 6B, 7A, 7B 8A, 8B, 9A, 9B, 10A, 10B, 11A, 11B, 11C, 12A, 12B, 12C, 12D, 13A, 13B, 13C, 14A, 14B, 15A, 15B, 16A, 16B, 17A, 17B, 18A, 18B, 19A, 19B, 20A, 20B, 21A, 21B, 21C, 21D, 21E, 22A, 22B, 23A, 23B, 23C, 24A, 24B, 24C, 25A, 25B, and 25C are cross-sectional views of intermediate stages in the manufacturing of nano-FETs, in accordance with some embodiments.

(4) FIGS. 26A, 26B, and 26C are cross-sectional views of a nano-FET, in accordance with some embodiments.

(5) FIGS. 27A, 27B, and 27C are cross-sectional views of a nano-FET, in accordance with some embodiments.

DETAILED DESCRIPTION

(6) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(7) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(8) In transistor gate stacks, a conductive material (e.g., tungsten) may be deposited as a fill metal with a chemical vapor deposition (CVD) process that uses a fluorine-comprising precursor. Depositing a fill metal in this manner may have advantages such as a relatively fast deposition process, which reduces manufacturing costs and increases yield. Various embodiments may further include a barrier layer between the fill metal and underlying work function metal (WFM) layers to prevent fluorine diffusion during deposition and/or the fill metal into the underlying WFM layers. For example, the barrier layer may comprise a tungsten layer that is deposited with fluorine free precursors. By preventing fluorine diffusion into the underlying layers, advantages may be achieved. For example, by preventing fluorine diffusion into the underlying layers, uneven fluorine distribution in the WFM layers may be avoided in different areas of the gate stack, which results in improved threshold voltage (V_t) stability and control. V_t stability and control may be especially improved in nano-FETs, which are susceptible to uneven fluorine diffusion in regions of the WFM layers between nanosheets compared to other regions of the WFM layers.

(9) FIG. 1 illustrates an example of nano-FETs (e.g., nanowire FETs, nanosheet FETs, or the like) in a three-dimensional view, in accordance with some embodiments. The nano-FETs comprise nano-structures 55 (e.g., nanosheets, nanowire, or the like) over fins 66 on a substrate 50 (e.g., a semiconductor substrate), wherein the nano-structures 55 act as channel regions for the nano-FETs. The nanostructure 55 may include p-type nanostructures, n-type nanostructures, or a combination thereof. Isolation regions 68 are disposed between adjacent fins 66, which may protrude above and from between neighboring isolation regions 68. Although the isolation regions 68 are described/illustrated as being separate from the substrate 50, as used herein, the term “substrate” may refer to the semiconductor substrate alone or a combination of the semiconductor substrate and the isolation regions. Additionally, although a bottom portion of the fins 66 are illustrated as being

single, continuous materials with the substrate **50**, the bottom portion of the fins **66** and/or the substrate **50** may comprise a single material or a plurality of materials. In this context, the fins **66** refer to the portion extending between the neighboring isolation regions **68**.

(10) Gate dielectrics **100** are over top surfaces of the fins **66** and along top surfaces, sidewalls, and bottom surfaces of the nano-structures **55**. Gate electrodes **102** are over the gate dielectrics **100**. Epitaxial source/drain regions **92** are disposed on the fins **66** on opposing sides of the gate dielectric layers **96** and the gate electrodes **98**.

(11) FIG. **1** further illustrates reference cross-sections that are used in later figures. Cross-section A-A' is along a longitudinal axis of a gate electrode **102** and in a direction, for example, perpendicular to the direction of current flow between the epitaxial source/drain regions **90** of a nano-FET. Cross-section B-B' is perpendicular to cross-section A-A' and is parallel to a longitudinal axis of a fin **66** of the nano-FET and in a direction of, for example, a current flow between the epitaxial source/drain regions **90** of the nano-FET. Cross-section C-C' is parallel to cross-section A-A' and extends through epitaxial source/drain regions of the nano-FETs. Subsequent figures refer to these reference cross-sections for clarity.

(12) Some embodiments discussed herein are discussed in the context of nano-FETs formed using a gate-last process. In other embodiments, a gate-first process may be used. Also, some embodiments contemplate aspects used in planar devices, such as planar FETs or in fin field-effect transistors (FinFETs).

(13) FIGS. **2** through **25C** are cross-sectional views of intermediate stages in the manufacturing of nano-FETs, in accordance with some embodiments. FIGS. **2** through **5**, **6A**, **13A**, **14A**, **15A**, **16A**, **17A**, **18A**, **19A**, **20A**, **21A**, **22A**, **23A**, **24A**, **25A**, **26A**, and **27A** illustrate reference cross-section A-A' illustrated in FIG. **1**. FIGS. **6B**, **7B**, **8B**, **9B**, **10B**, **11B**, **11C**, **12B**, **12D**, **13B**, **14B**, **15B**, **16B**, **17B**, **18B**, **19B**, **20B**, **21B**, **22B**, **23B**, **24B**, **25B**, **26B**, and **27B** illustrate reference cross-section B-B' illustrated in FIG. **1**. FIGS. **7A**, **8A**, **9A**, **10A**, **11A**, **12A**, **12C**, **13C**, **23C**, **24C**, **25C**, **26C**, and **27C** illustrate reference cross-section C-C' illustrated in FIG. **1**.

(14) In FIG. **2**, a substrate **50** is provided. The substrate **50** may be a semiconductor substrate, such as a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like, which may be doped (e.g., with a p-type or an n-type dopant) or undoped. The substrate **50** may be a wafer, such as a silicon wafer. Generally, an SOI substrate is a layer of a semiconductor material formed on an insulator layer. The insulator layer may be, for example, a buried oxide (BOX) layer, a silicon oxide layer, or the like. The insulator layer is provided on a substrate, typically a silicon or glass substrate. Other substrates, such as a multi-layered or gradient substrate may also be used. In some embodiments, the semiconductor material of the substrate **50** may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including silicon-germanium, gallium arsenide phosphide, aluminum indium arsenide, aluminum gallium arsenide, gallium indium arsenide, gallium indium phosphide, and/or gallium indium arsenide phosphide; or combinations thereof.

(15) The substrate **50** has an n-type region **50N** and a p-type region **50P**. The n-type region **50N** can be for forming n-type devices, such as NMOS transistors, e.g., n-type nano-FETs, and the p-type region **50P** can be for forming p-type devices, such as PMOS transistors, e.g., p-type nano-FETs. The n-type region **50N** may be physically separated from the p-type region **50P** (as illustrated by divider **20**), and any number of device features (e.g., other active devices, doped regions, isolation structures, etc.) may be disposed between the n-type region **50N** and the p-type region **50P**. Although one n-type region **50N** and one p-type region **50P** are illustrated, any number of n-type regions **50N** and p-type regions **50P** may be provided. In some embodiments, one or more wells and/or an anti-punch through (APT) layer may be formed in the substrate **50** through one or more suitable implantation steps.

(16) Further in FIG. **2**, a multi-layer stack **64** is formed over the substrate **50**. The multi-layer stack

64 includes alternating layers of first semiconductor layers **51A-C** (collectively referred to as first semiconductor layers **51**) and second semiconductor layers **53A-C** (collectively referred to as second semiconductor layers **53**). For purposes of illustration and as discussed in greater detail below, the second semiconductor layers **53** will be removed and the first semiconductor layers **51** will be patterned to form channel regions of nano-FETs in the p-type region **50P**. Also, the first semiconductor layers **51** will be removed and the second semiconductor layers **53** will be patterned to form channel regions of nano-FETs in the n-type regions **50N**. Nevertheless, in some embodiments the first semiconductor layers **51** may be removed and the second semiconductor layers **53** may be patterned to form channel regions of nano-FETs in the n-type region **50N**, and the second semiconductor layers **53** may be removed and the first semiconductor layers **51** may be patterned to form channel regions of nano-FETs in the p-type regions **50P**.

(17) In still other embodiments, the first semiconductor layers **51** may be removed and the second semiconductor layers **53** may be patterned to form channel regions of nano-FETs in both the n-type region **50N** and the p-type region **50P**. In other embodiments, the second semiconductor layers **53** may be removed and the first semiconductor layers **51** may be patterned to form channel regions of non-FETs in both the n-type region **50N** and the p-type region **50P**. In such embodiments, the channel regions in both the n-type region **50N** and the p-type region **50P** may have a same material composition (e.g., silicon, or the like) and be formed simultaneously. FIGS. **26A**, **26B**, and **27C** illustrate a structure resulting from such embodiments where the channel regions in both the p-type region **50P** and the n-type region **50N** comprise silicon, for example.

(18) The multi-layer stack **64** is illustrated as including three layers of each of the first semiconductor layers **51** and the second semiconductor layers **53** for illustrative purposes. In some embodiments, the multi-layer stack **64** may include any number of the first semiconductor layers **51** and the second semiconductor layers **53**. Each of the layers of the multi-layer stack **64** may be epitaxially grown using a process such as chemical vapor deposition (CVD), atomic layer deposition (ALD), vapor phase epitaxy (VPE), molecular beam epitaxy (MBE), or the like. In various embodiments, the first semiconductor layers **51** may be formed of a first semiconductor material suitable for p-type nano-FETs, such as silicon germanium, or the like, and the second semiconductor layers **53** may be formed of a second semiconductor material suitable for n-type nano-FETs, such as silicon, silicon carbon, or the like. The multi-layer stack **64** is illustrated as having a bottommost semiconductor layer suitable for p-type nano-FETs for illustrative purposes. In some embodiments, multi-layer stack **64** may be formed such that the bottommost layer is a semiconductor layer suitable for n-type nano-FETs.

(19) The first semiconductor materials and the second semiconductor materials may be materials having a high-etch selectivity to one another. As such, the first semiconductor layers **51** of the first semiconductor material may be removed without significantly removing the second semiconductor layers **53** of the second semiconductor material in the n-type region **50N**, thereby allowing the second semiconductor layers **53** to be patterned to form channel regions of n-type NSFETs. Similarly, the second semiconductor layers **53** of the second semiconductor material may be removed without significantly removing the first semiconductor layers **51** of the first semiconductor material in the p-type region **50P**, thereby allowing the first semiconductor layers **51** to be patterned to form channel regions of p-type NSFETs. In other embodiments, the channel regions in the n-type region **50N** and the p-type region **50P** may be formed simultaneously and have a same material composition, such as silicon, silicon germanium, or the like. FIGS. **26A**, **26B**, and **27C** illustrate a structure resulting from such embodiments where the channel regions in both the p-type region **50P** and the n-type region **50N** comprise silicon, for example.

(20) Referring now to FIG. **3**, fins **66** are formed in the substrate **50** and nanostructures **55** are formed in the multi-layer stack **64**, in accordance with some embodiments. In some embodiments, the nanostructures **55** and the fins **66** may be formed in the multi-layer stack **64** and the substrate **50**, respectively, by etching trenches in the multi-layer stack **64** and the substrate **50**. The etching

may be any acceptable etch process, such as a reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etching may be anisotropic. Forming the nanostructures **55** by etching the multi-layer stack **64** may further define first nanostructures **52A-C** (collectively referred to as the first nanostructures **52**) from the first semiconductor layers **51** and define second nanostructures **54A-C** (collectively referred to as the second nanostructures **54**) from the second semiconductor layers **53**. The first nanostructures **52** and the second nanostructures **54** may further be collectively referred to as nanostructures **55**.

(21) The fins **66** and the nanostructures **55** may be patterned by any suitable method. For example, the fins **66** and the nanostructures **55** may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins **66**.

(22) FIG. **3** illustrates the fins **66** in the n-type region **50N** and the p-type region **50P** as having substantially equal widths for illustrative purposes. In some embodiments, widths of the fins **66** in the n-type region **50N** may be greater or thinner than the fins **66** in the p-type region **50P**. Further, while each of the fins **66** and the nanostructures **55** are illustrated as having a consistent width throughout, in other embodiments, the fins **66** and/or the nanostructures **55** may have tapered sidewalls such that a width of each of the fins **66** and/or the nanostructures **55** continuously increases in a direction towards the substrate **50**. In such embodiments, each of the nanostructures **55** may have a different width and be trapezoidal in shape.

(23) In FIG. **4**, shallow trench isolation (STI) regions **68** are formed adjacent the fins **66**. The STI regions **68** may be formed by depositing an insulation material over the substrate **50**, the fins **66**, and nanostructures **55**, and between adjacent fins **66**. The insulation material may be an oxide, such as silicon oxide, a nitride, the like, or a combination thereof, and may be formed by high-density plasma CVD (HDP-CVD), flowable CVD (FCVD), the like, or a combination thereof. Other insulation materials formed by any acceptable process may be used. In the illustrated embodiment, the insulation material is silicon oxide formed by an FCVD process. An anneal process may be performed once the insulation material is formed. In an embodiment, the insulation material is formed such that excess insulation material covers the nanostructures **55**. Although the insulation material is illustrated as a single layer, some embodiments may utilize multiple layers. For example, in some embodiments a liner (not separately illustrated) may first be formed along a surface of the substrate **50**, the fins **66**, and the nanostructures **55**. Thereafter, a fill material, such as those discussed above may be formed over the liner.

(24) A removal process is then applied to the insulation material to remove excess insulation material over the nanostructures **55**. In some embodiments, a planarization process such as a chemical mechanical polish (CMP), an etch-back process, combinations thereof, or the like may be utilized. The planarization process exposes the nanostructures **55** such that top surfaces of the nanostructures **55** and the insulation material are level after the planarization process is complete.

(25) The insulation material is then recessed to form the STI regions **68**. The insulation material is recessed such that upper portions of fins **66** in the regions **50N** and the region **50P** protrude from between neighboring STI regions **68**. Further, the top surfaces of the STI regions **68** may have a flat surface as illustrated, a convex surface, a concave surface (such as dishing), or a combination thereof. The top surfaces of the STI regions **68** may be formed flat, convex, and/or concave by an appropriate etch. The STI regions **68** may be recessed using an acceptable etching process, such as one that is selective to the material of the insulation material (e.g., etches the material of the insulation material at a faster rate than the material of the fins **66** and the nanostructures **55**). For

example, an oxide removal using, for example, dilute hydrofluoric (dHF) acid may be used.

(26) The process described above with respect to FIGS. 2 through 4 is just one example of how the fins 66 and the nanostructures 55 may be formed. In some embodiments, the fins 66 and/or the nanostructures 55 may be formed using a mask and an epitaxial growth process. For example, a dielectric layer can be formed over a top surface of the substrate 50, and trenches can be etched through the dielectric layer to expose the underlying substrate 50. Epitaxial structures can be epitaxially grown in the trenches, and the dielectric layer can be recessed such that the epitaxial structures protrude from the dielectric layer to form the fins 66 and/or the nanostructures 55. The epitaxial structures may comprise the alternating semiconductor materials discussed above, such as the first semiconductor materials and the second semiconductor materials. In some embodiments where epitaxial structures are epitaxially grown, the epitaxially grown materials may be in situ doped during growth, which may obviate prior and/or subsequent implantations, although in situ and implantation doping may be used together.

(27) Additionally, the first semiconductor layers 51 (and resulting nanostructures 52) and the second semiconductor layers 53 (and resulting nanostructures 54) are illustrated and discussed herein as comprising the same materials in the p-type region 50P and the n-type region 50N for illustrative purposes only. As such, in some embodiments one or both of the first semiconductor layers 51 and the second semiconductor layers 53 may be different materials or formed in a different order in the p-type region 50P and the n-type region 50N.

(28) Further in FIG. 4, appropriate wells (not separately illustrated) may be formed in the fins 66, the nanostructures 55, and/or the STI regions 68. In embodiments with different well types, different implant steps for the n-type region 50N and the p-type region 50P may be achieved using a photoresist or other masks (not separately illustrated). For example, a photoresist may be formed over the fins 66 and the STI regions 68 in the n-type region 50N and the p-type region 50P. The photoresist is patterned to expose the p-type region 50P. The photoresist can be formed by using a spin-on technique and can be patterned using acceptable photolithography techniques. Once the photoresist is patterned, an n-type impurity implant is performed in the p-type region 50P, and the photoresist may act as a mask to substantially prevent n-type impurities from being implanted into the n-type region 50N. The n-type impurities may be phosphorus, arsenic, antimony, or the like implanted in the region to a concentration in a range from about 10^{13} atoms/cm³ to about 10^{14} atoms/cm³. After the implant, the photoresist is removed, such as by an acceptable ashing process.

(29) Following or prior to the implanting of the p-type region 50P, a photoresist or other masks (not separately illustrated) is formed over the fins 66, the nanostructures 55, and the STI regions 68 in the p-type region 50P and the n-type region 50N. The photoresist is patterned to expose the n-type region 50N. The photoresist can be formed by using a spin-on technique and can be patterned using acceptable photolithography techniques. Once the photoresist is patterned, a p-type impurity implant may be performed in the n-type region 50N, and the photoresist may act as a mask to substantially prevent p-type impurities from being implanted into the p-type region 50P. The p-type impurities may be boron, boron fluoride, indium, or the like implanted in the region to a concentration in a range from about 10^{13} atoms/cm³ to about 10^{14} atoms/cm³. After the implant, the photoresist may be removed, such as by an acceptable ashing process.

(30) After the implants of the n-type region 50N and the p-type region 50P, an anneal may be performed to repair implant damage and to activate the p-type and/or n-type impurities that were implanted. In some embodiments, the grown materials of epitaxial fins may be in situ doped during growth, which may obviate the implantations, although in situ and implantation doping may be used together.

(31) In FIG. 5, a dummy dielectric layer 70 is formed on the fins 66 and/or the nanostructures 55. The dummy dielectric layer 70 may be, for example, silicon oxide, silicon nitride, a combination thereof, or the like, and may be deposited or thermally grown according to acceptable techniques. A

dummy gate layer 72 is formed over the dummy dielectric layer 70, and a mask layer 74 is formed over the dummy gate layer 72. The dummy gate layer 72 may be deposited over the dummy dielectric layer 70 and then planarized, such as by a CMP. The mask layer 74 may be deposited over the dummy gate layer 72. The dummy gate layer 72 may be a conductive or non-conductive material and may be selected from a group including amorphous silicon, polycrystalline-silicon (polysilicon), poly-crystalline silicon-germanium (poly-SiGe), metallic nitrides, metallic silicides, metallic oxides, and metals. The dummy gate layer 72 may be deposited by physical vapor deposition (PVD), CVD, sputter deposition, or other techniques for depositing the selected material. The dummy gate layer 72 may be made of other materials that have a high etching selectivity from the etching of isolation regions. The mask layer 74 may include, for example, silicon nitride, silicon oxynitride, or the like. In this example, a single dummy gate layer 72 and a single mask layer 74 are formed across the n-type region 50N and the p-type region 50P. It is noted that the dummy dielectric layer 70 is shown covering only the fins 66 and the nanostructures 55 for illustrative purposes only. In some embodiments, the dummy dielectric layer 70 may be deposited such that the dummy dielectric layer 70 covers the STI regions 68, such that the dummy dielectric layer 70 extends between the dummy gate layer 72 and the STI regions 68.

(32) FIGS. 6A through 16B illustrate various additional steps in the manufacturing of embodiment devices. FIGS. 7A, 8A, 9A, 10A, 11A, 12A, 12C, 13A, 13C, 14A, and 15A illustrate features in either the regions 50N or the regions 50P. In FIGS. 6A and 6B, the mask layer 74 (see FIG. 5) may be patterned using acceptable photolithography and etching techniques to form masks 78. The pattern of the masks 78 then may be transferred to the dummy gate layer 72 and to the dummy dielectric layer 70 to form dummy gates 76 and dummy gate dielectrics 71, respectively. The dummy gates 76 cover respective channel regions of the fins 66. The pattern of the masks 78 may be used to physically separate each of the dummy gates 76 from adjacent dummy gates 76. The dummy gates 76 may also have a lengthwise direction substantially perpendicular to the lengthwise direction of respective fins 66.

(33) In FIGS. 7A and 7B, a first spacer layer 80 and a second spacer layer 82 are formed over the structures illustrated in FIGS. 6A and 6B, respectively. The first spacer layer 80 and the second spacer layer 82 will be subsequently patterned to act as spacers for forming self-aligned source/drain regions. In FIGS. 7A and 7B, the first spacer layer 80 is formed on top surfaces of the STI regions 68; top surfaces and sidewalls of the fins 66, the nanostructures 55, and the masks 78; and sidewalls of the dummy gates 76 and the dummy gate dielectric 71. The second spacer layer 82 is deposited over the first spacer layer 80. The first spacer layer 80 may be formed of silicon oxide, silicon nitride, silicon oxynitride, or the like, using techniques such as thermal oxidation or deposited by CVD, ALD, or the like. The second spacer layer 82 may be formed of a material having a different etch rate than the material of the first spacer layer 80, such as silicon oxide, silicon nitride, silicon oxynitride, or the like, and may be deposited by CVD, ALD, or the like.

(34) After the first spacer layer 80 is formed and prior to forming the second spacer layer 82, implants for lightly doped source/drain (LDD) regions (not separately illustrated) may be performed. In embodiments with different device types, similar to the implants discussed above in FIG. 4, a mask, such as a photoresist, may be formed over the n-type region 50N, while exposing the p-type region 50P, and appropriate type (e.g., p-type) impurities may be implanted into the exposed fins 66 and nanostructures 55 in the p-type region 50P. The mask may then be removed. Subsequently, a mask, such as a photoresist, may be formed over the p-type region 50P while exposing the n-type region 50N, and appropriate type impurities (e.g., n-type) may be implanted into the exposed fins 66 and nanostructures 55 in the n-type region 50N. The mask may then be removed. The n-type impurities may be the any of the n-type impurities previously discussed, and the p-type impurities may be the any of the p-type impurities previously discussed. The lightly doped source/drain regions may have a concentration of impurities in a range from about 1×10^{15} atoms/cm³ to about 1×10^{19} atoms/cm³. An anneal may be used to

repair implant damage and to activate the implanted impurities.

(35) In FIGS. **8A** and **8B**, the first spacer layer **80** and the second spacer layer **82** are etched to form first spacers **81** and second spacers **83**. As will be discussed in greater detail below, the first spacers **81** and the second spacers **83** act to self-aligned subsequently formed source drain regions, as well as to protect sidewalls of the fins **66** and/or nanostructure **55** during subsequent processing. The first spacer layer **80** and the second spacer layer **82** may be etched using a suitable etching process, such as an isotropic etching process (e.g., a wet etching process), an anisotropic etching process (e.g., a dry etching process), or the like. In some embodiments, the material of the second spacer layer **82** has a different etch rate than the material of the first spacer layer **80**, such that the first spacer layer **80** may act as an etch stop layer when patterning the second spacer layer **82** and such that the second spacer layer **82** may act as a mask when patterning the first spacer layer **80**. For example, the second spacer layer **82** may be etched using an anisotropic etch process wherein the first spacer layer **80** acts as an etch stop layer, wherein remaining portions of the second spacer layer **82** form second spacers **83** as illustrated in FIG. **8A**. Thereafter, the second spacers **83** acts as a mask while etching exposed portions of the first spacer layer **80**, thereby forming first spacers **81** as illustrated in FIG. **8A**.

(36) As illustrated in FIG. **8A**, the first spacers **81** and the second spacers **83** are disposed on sidewalls of the fins **66** and/or nanostructures **55**. As illustrated in FIG. **8B**, in some embodiments, the second spacer layer **82** may be removed from over the first spacer layer **80** adjacent the masks **78**, the dummy gates **76**, and the dummy gate dielectrics **71**, and the first spacers **81** are disposed on sidewalls of the masks **78**, the dummy gates **76**, and the dummy dielectric layers **60**. In other embodiments, a portion of the second spacer layer **82** may remain over the first spacer layer **80** adjacent the masks **78**, the dummy gates **76**, and the dummy gate dielectrics **71**.

(37) It is noted that the above disclosure generally describes a process of forming spacers and LDD regions. Other processes and sequences may be used. For example, fewer or additional spacers may be utilized, different sequence of steps may be utilized (e.g., the first spacers **81** may be patterned prior to depositing the second spacer layer **82**), additional spacers may be formed and removed, and/or the like. Furthermore, the n-type and p-type devices may be formed using different structures and steps.

(38) In FIGS. **9A** and **9B**, first recesses **86** are formed in the fins **66**, the nanostructures **55**, and the substrate **50**, in accordance with some embodiments. Epitaxial source/drain regions will be subsequently formed in the first recesses **86**. The first recesses **86** may extend through the first nanostructures **52** and the second nanostructures **54**, and into the substrate **50**. As illustrated in FIG. **9A**, top surfaces of the STI regions **58** may be level with bottom surfaces of the first recesses **86**. In various embodiments, the fins **66** may be etched such that bottom surfaces of the first recesses **86** are disposed below the top surfaces of the STI regions **68**; or the like. The first recesses **86** may be formed by etching the fins **66**, the nanostructures **55**, and the substrate **50** using anisotropic etching processes, such as RIE, NBE, or the like. The first spacers **81**, the second spacers **83**, and the masks **78** mask portions of the fins **66**, the nanostructures **55**, and the substrate **50** during the etching processes used to form the first recesses **86**. A single etch process or multiple etch processes may be used to etch each layer of the nanostructures **55** and/or the fins **66**. Timed etch processes may be used to stop the etching of the first recesses **86** after the first recesses **86** reach a desired depth.

(39) In FIGS. **10A** and **10B**, portions of sidewalls of the layers of the multi-layer stack **64** formed of the first semiconductor materials (e.g., the first nanostructures **52**) exposed by the first recesses **86** are etched to form sidewall recesses **88** in the n-type region **50N**, and portions of sidewalls of the layers of the multi-layer stack **56** formed of the second semiconductor materials (e.g., the second nanostructures **54**) exposed by the first recesses **86** are etched to form sidewall recesses **88** in the p-type region **50N**. Although sidewalls of the first nanostructures **52** and the second nanostructures **54** in recesses **88** are illustrated as being straight in FIG. **10B**, the sidewalls may be concave or convex. The sidewalls may be etched using isotropic etching processes, such as wet

etching or the like. The p-type region **50P** may be protected using a mask (not shown) while etchants selective to the first semiconductor materials are used to etch the first nanostructures **52** such that the second nanostructures **54** and the substrate **50** remain relatively unetched as compared to the first nanostructures **52** in the n-type region **50N**. Similarly, the n-type region **50N** may be protected using a mask (not shown) while etchants selective to the second semiconductor materials are used to etch the second nanostructures **54** such that the first nanostructures **52** and the substrate **50** remain relatively unetched as compared to the second nanostructures **54** in the p-type region **50P**. In an embodiment in which the first nanostructures **52** include, e.g., SiGe, and the second nanostructures **54** include, e.g., Si or SiC, a dry etch process with tetramethylammonium hydroxide (TMAH), ammonium hydroxide (NH₄OH), or the like may be used to etch sidewalls of the first nanostructures **52** in the n-type region **50N**, and a dry etch process with hydrogen fluoride, another fluorine-based gas, or the like may be used to etch sidewalls of the second nanostructures **54** in the p-type region **50P**.

(40) In FIGS. **11A-11C**, first inner spacers **90** are formed in the sidewall recess **88**. The first inner spacers **90** may be formed by depositing an inner spacer layer (not separately illustrated) over the structures illustrated in FIGS. **10A** and **10B**. The first inner spacers **90** act as isolation features between subsequently formed source/drain regions and a gate structure. As will be discussed in greater detail below, source/drain regions will be formed in the recesses **86**, while the first nanostructures **52** in the n-type region **50N** and the second nanostructures **54** in the p-type region **50P** will be replaced with corresponding gate structures.

(41) The inner spacer layer may be deposited by a conformal deposition process, such as CVD, ALD, or the like. The inner spacer layer may comprise a material such as silicon nitride or silicon oxynitride, although any suitable material, such as low-dielectric constant (low-k) materials having a k-value less than about 3.5, may be utilized. The inner spacer layer may then be anisotropically etched to form the first inner spacers **90**. Although outer sidewalls of the first inner spacers **90** are illustrated as being flush with sidewalls of the second nanostructures **54** in the n-type region **50N** and flush with the sidewalls of the first nanostructures **52** in the p-type region **50P**, the outer sidewalls of the first inner spacers **90** may extend beyond or be recessed from sidewalls of the second nanostructures **54** and/or the first nanostructures **52**, respectively.

(42) Moreover, although the outer sidewalls of the first inner spacers **90** are illustrated as being straight in FIG. **11B**, the outer sidewalls of the first inner spacers **90** may be concave or convex. As an example, FIG. **11C** illustrates an embodiment in which sidewalls of the first nanostructures **52** are concave, outer sidewalls of the first inner spacers **90** are concave, and the first inner spacers are recessed from sidewalls of the second nanostructures **54** in the n-type region **50N**. Also illustrated are embodiments in which sidewalls of the second nanostructures **54** are concave, outer sidewalls of the first inner spacers **90** are concave, and the first inner spacers are recessed from sidewalls of the first nanostructures **52** in the p-type region **50P**. The inner spacer layer may be etched by an anisotropic etching process, such as RIE, NBE, or the like. The first inner spacers **90** may be used to prevent damage to subsequently formed source/drain regions (such as the epitaxial source/drain regions **92**, discussed below with respect to FIGS. **12A-12C**) by subsequent etching processes, such as etching processes used to form gate structures.

(43) In FIGS. **12A-12C**, epitaxial source/drain regions **92** are formed in the first recesses **86**. In some embodiments, the source/drain regions **92** may exert stress on the second nanostructures **54** in the n-type region **50N** and on the first nanostructures **52** in the p-type region **50P**, thereby improving performance. As illustrated in FIG. **12B**, the epitaxial source/drain regions **92** are formed in the first recesses **86** such that each dummy gate **76** is disposed between respective neighboring pairs of the epitaxial source/drain regions **92**. In some embodiments, the first spacers **81** are used to separate the epitaxial source/drain regions **92** from the dummy gates **72** and the first inner spacers **90** are used to separate the epitaxial source/drain regions **92** from the nanostructures **55** by an appropriate lateral distance so that the epitaxial source/drain regions **92** do not short out

with subsequently formed gates of the resulting nano-FETs.

(44) The epitaxial source/drain regions **92** in the n-type region **50N**, e.g., the NMOS region, may be formed by masking the p-type region **50P**, e.g., the PMOS region. Then, the epitaxial source/drain regions **92** are epitaxially grown in the first recesses **86** in the n-type region **50N**. The epitaxial source/drain regions **92** may include any acceptable material appropriate for n-type nano-FETs. For example, if the second nanostructures **54** are silicon, the epitaxial source/drain regions **92** may include materials exerting a tensile strain on the second nanostructures **54**, such as silicon, silicon carbide, phosphorous doped silicon carbide, silicon phosphide, or the like. The epitaxial source/drain regions **92** may have surfaces raised from respective upper surfaces of the nanostructures **55** and may have facets.

(45) The epitaxial source/drain regions **92** in the p-type region **50P**, e.g., the PMOS region, may be formed by masking the n-type region **50N**, e.g., the NMOS region. Then, the epitaxial source/drain regions **92** are epitaxially grown in the first recesses **86** in the p-type region **50P**. The epitaxial source/drain regions **92** may include any acceptable material appropriate for p-type nano-FETs. For example, if the first nanostructures **52** are silicon germanium, the epitaxial source/drain regions **92** may comprise materials exerting a compressive strain on the first nanostructures **52**, such as silicon-germanium, boron doped silicon-germanium, germanium, germanium tin, or the like. The epitaxial source/drain regions **92** may also have surfaces raised from respective surfaces of the multi-layer stack **56** and may have facets.

(46) The epitaxial source/drain regions **92**, the first nanostructures **52**, the second nanostructures **54**, and/or the substrate **50** may be implanted with dopants to form source/drain regions, similar to the process previously discussed for forming lightly-doped source/drain regions, followed by an anneal. The source/drain regions may have an impurity concentration of between about 1×10^{19} atoms/cm³ and about 1×10^{21} atoms/cm³. The n-type and/or p-type impurities for source/drain regions may be any of the impurities previously discussed. In some embodiments, the epitaxial source/drain regions **92** may be in situ doped during growth.

(47) As a result of the epitaxy processes used to form the epitaxial source/drain regions **92** in the n-type region **50N** and the p-type region **50P**, upper surfaces of the epitaxial source/drain regions **92** have facets which expand laterally outward beyond sidewalls of the nanostructures **55**. In some embodiments, these facets cause adjacent epitaxial source/drain regions **92** of a same NSFET to merge as illustrated by FIG. **12A**. In other embodiments, adjacent epitaxial source/drain regions **92** remain separated after the epitaxy process is completed as illustrated by FIG. **12C**. In the embodiments illustrated in FIGS. **12A** and **12C**, the first spacers **81** may be formed to a top surface of the STI regions **68** thereby blocking the epitaxial growth. In some other embodiments, the first spacers **81** may cover portions of the sidewalls of the nanostructures **55** further blocking the epitaxial growth. In some other embodiments, the spacer etch used to form the first spacers **81** may be adjusted to remove the spacer material to allow the epitaxially grown region to extend to the surface of the STI region **58**.

(48) The epitaxial source/drain regions **92** may comprise one or more semiconductor material layers. For example, the epitaxial source/drain regions **92** may comprise a first semiconductor material layer **92A**, a second semiconductor material layer **92B**, and a third semiconductor material layer **92C**. Any number of semiconductor material layers may be used for the epitaxial source/drain regions **92**. Each of the first semiconductor material layer **92A**, the second semiconductor material layer **92B**, and the third semiconductor material layer **92C** may be formed of different semiconductor materials and may be doped to different dopant concentrations. In some embodiments, the first semiconductor material layer **92A** may have a dopant concentration less than the second semiconductor material layer **92B** and greater than the third semiconductor material layer **92C**. In embodiments in which the epitaxial source/drain regions **92** comprise three semiconductor material layers, the first semiconductor material layer **92A** may be deposited, the second semiconductor material layer **92B** may be deposited over the first semiconductor material

layer **92A**, and the third semiconductor material layer **92C** may be deposited over the second semiconductor material layer **92B**.

(49) FIG. **12D** illustrates an embodiment in which sidewalls of the first nanostructures **52** in the n-type region **50N** and sidewalls of the second nanostructures **54** in the p-type region **50P** are concave, outer sidewalls of the first inner spacers **90** are concave, and the first inner spacers **90** are recessed from sidewalls of the second nanostructures **54** and the first nanostructures **52**, respectively. As illustrated in FIG. **12D**, the epitaxial source/drain regions **92** may be formed in contact with the first inner spacers **90** and may extend past sidewalls of the second nanostructures **54** in the n-type region **50N** and past sidewalls of the first nanostructures **52** in the p-type region **50P**. Further, in embodiments where the first inner spacers **90** are recessed from sidewalls of the second nanostructures **54** and/or the first nanostructures **52**, the epitaxial source/drain regions **92** may be formed between the second nanostructures **54** and/or the first nanostructures **52**, respectively.

(50) In FIGS. **13A-13C**, a first interlayer dielectric (ILD) **96** is deposited over the structure illustrated in FIGS. **6A**, **12B**, and **12A** (the processes of FIGS. **7A-12D** do not alter the cross-section illustrated in FIGS. **6A**), respectively. The first ILD **96** may be formed of a dielectric material, and may be deposited by any suitable method, such as CVD, plasma-enhanced CVD (PECVD), or FCVD. Dielectric materials may include phospho-silicate glass (PSG), boro-silicate glass (BSG), boron-doped phospho-silicate glass (BPSG), undoped silicate glass (USG), or the like. Other insulation materials formed by any acceptable process may be used. In some embodiments, a contact etch stop layer (CESL) **94** is disposed between the first ILD **96** and the epitaxial source/drain regions **92**, the masks **74**, and the first spacers **81**. The CESL **94** may comprise a dielectric material, such as, silicon nitride, silicon oxide, silicon oxynitride, or the like, having a different etch rate than the material of the overlying first ILD **96**.

(51) In FIGS. **14A-14C**, a planarization process, such as a CMP, may be performed to level the top surface of the first ILD **96** with the top surfaces of the dummy gates **76** or the masks **78**. The planarization process may also remove the masks **78** on the dummy gates **76**, and portions of the first spacers **81** along sidewalls of the masks **78**. After the planarization process, top surfaces of the dummy gates **76**, the first spacers **81**, and the first ILD **96** are level within process variations. Accordingly, the top surfaces of the dummy gates **72** are exposed through the first ILD **96**. In some embodiments, the masks **78** may remain, in which case the planarization process levels the top surface of the first ILD **96** with top surface of the masks **78** and the first spacers **81**.

(52) In FIGS. **15A** and **15B**, the dummy gates **72**, and the masks **74** if present, are removed in one or more etching steps, so that second recesses **98** are formed. Portions of the dummy gate dielectrics **60** in the second recesses **98** are also be removed. In some embodiments, the dummy gates **72** and the dummy gate dielectrics **60** are removed by an anisotropic dry etch process. For example, the etching process may include a dry etch process using reaction gas(es) that selectively etch the dummy gates **72** at a faster rate than the first ILD **96** or the first spacers **81**. Each second recess **98** exposes and/or overlies portions of nanostructures **55**, which act as channel regions in subsequently completed nano-FETs. Portions of the nanostructures **55** which act as the channel regions are disposed between neighboring pairs of the epitaxial source/drain regions **92**. During the removal, the dummy dielectric layers **60** may be used as etch stop layers when the dummy gates **72** are etched. The dummy dielectric layers **60** may then be removed after the removal of the dummy gates **72**.

(53) In FIGS. **16A** and **16B**, the second nanostructures **54** in the p-type region **50P** may be removed by forming a mask (not shown) over the n-type region **50N** and performing an isotropic etching process such as wet etching or the like using etchants which are selective to the materials of the second nanostructures **54**, while the first nanostructures **52**, the substrate **50**, the STI regions **58** remain relatively unetched as compared to the second nanostructures **54**. In embodiments in which the second nanostructures **54** include, e.g., SiGe, and the first nanostructures **52** include, e.g., Si or

SiC, hydrogen fluoride, another fluorine-based gas, or the like may be used to remove the second nanostructures **54** in the p-type region **50P**.

(54) In other embodiments, the channel regions in the n-type region **50N** and the p-type region **50P** may be formed simultaneously, for example by removing the first nanostructures **52** in both the n-type region **50N** and the p-type region **50P** or by removing the second nanostructures **54** in both the n-type region **50N** and the p-type region **50P**. In such embodiments, channel regions of n-type NSFETs and p-type NSFETs may have a same material composition, such as silicon, silicon germanium, or the like. FIGS. **26A** and **26B** illustrate a structure resulting from such embodiments where the channel regions in both the p-type region **50P** and the n-type region **50N** are provided by the second nanostructures **54** and comprise silicon, for example.

(55) In FIGS. **17A** through **22B** gate dielectric layers and gate electrodes are formed for replacement gates in the second recesses **98** according to some embodiments. The gate electrodes include a barrier layer between the WFM layer(s) and a gate fill metal to prevent (or at least reduce) fluorine diffusion from the fill metal into the underlying WFM layers. As a result, uneven diffusion of fluorine in the WFM layer(s) can be avoided, and threshold voltage stability and control in the gates may be improved.

(56) The formation of the gate dielectrics in the n-type region **50N** and the p-type region **50P** may occur simultaneously such that the gate dielectrics in each region are formed from the same materials, and the formation of the gate electrodes may occur simultaneously such that the gate electrodes in each region are formed from the same materials. In some embodiments, the gate dielectrics in each region may be formed by distinct processes, such that the gate dielectrics may be different materials and/or have a different number of layers, and/or the gate electrodes in each region may be formed by distinct processes, such that the gate electrodes may be different materials and/or have a different number of layers. Various masking steps may be used to mask and expose appropriate regions when using distinct processes. In the following description, the gate electrodes of the n-type region **50N** and the gate electrodes of the p-type region **50P** are formed separately.

(57) FIGS. **17A** through **21B** illustrate forming the gate dielectrics **100** and the gate electrodes **102** in the p-type region **50P**, and the n-type region **50N** may be masked at least while forming the gate electrodes **102** in the p-type region **50P** (e.g., as described below in FIGS. **18A** through **21B**).

(58) In FIGS. **17A** and **17B**, gate dielectrics **100** are deposited conformally in the second recesses **98** in the p-type region **50P**. The gate dielectrics **100** comprise one or more dielectric layers, such as an oxide, a metal oxide, the like, or combinations thereof. For example, in some embodiments, the gate dielectrics **100** may comprise a first gate dielectric **101** (e.g., comprising silicon oxide, or the like) and a second gate dielectric **103** (e.g., comprising a metal oxide, or the like) over the first gate dielectric **101**. In some embodiments, the second gate dielectric **103** includes a high-k dielectric material, and in these embodiments, the second gate dielectric **103** may have a k value greater than about 7.0, and may include a metal oxide or a silicate of hafnium, aluminum, zirconium, lanthanum, manganese, barium, titanium, lead, and combinations thereof. The first gate dielectric **101** may be referred to as an interfacial layer, and the second gate dielectric **103** may be referred to as a high-k gate dielectric in some embodiments.

(59) The structure of the gate dielectrics **100** may be the same or different in the n-type region **50N** and the p-type region **50P**. For example, the n-type region **50N** may be masked or exposed while forming the gate dielectrics **100** in the p-type region **50P**. In embodiments where the n-type region **50N** is exposed, the gate dielectrics **100** may be simultaneously formed in the n-type regions **50N**. The formation methods of the gate dielectrics **100** may include molecular-beam deposition (MBD), ALD, PECVD, and the like.

(60) In FIGS. **18A** and **18B**, a conductive material **105** is deposited conformally on gate dielectrics **100** in the p-type region **50P**. In some embodiments, the conductive material **105** is a p-type WFM, comprising titanium nitride, tantalum nitride, tungsten nitride, molybdenum nitride, combinations thereof, or the like. In some embodiments, the conductive material **105** has a multilayer structure

and include more than one conductive material. For example, the conductive material **105** may comprise a layer of titanium nitride and a layer of a tantalum nitride on the layer of titanium nitride. The conductive material **105** may be deposited by CVD, ALD, PECVD, PVD, or the like. In some embodiments, the conductive material **105** may be deposited on exposed surfaces of the gate dielectrics **100** to a sufficient thickness such that the conductive material **105** merges in regions **501** between adjacent first nanostructures **52** (e.g., the first nanostructures **52A**, **52B**, and **52C**). For example, the conductive material **105** may be deposited on surfaces of the first nanostructures **52** in the regions **501**, and as the conductive material **105** increases in thickness during the deposition, separated portions of the conductive material **105** may touch and merge along seams **105S**.

Specifically, deposition of the conductive material **105** may continue until a first portion **105A** of the conductive material **105** merges with a second portion **105B** of the conductive material **105** in the regions **501**.

(61) In FIGS. **19A** and **19B** an adhesion layer **107** may be deposited over the conductive material **105**. In some embodiments, the adhesion layer **107** is deposited conformally on the conductive material **105** in the p-type region **50P**. In some embodiments, the adhesion layer **107** comprises titanium nitride, tantalum nitride, or the like. The adhesion layer **107** may be deposited by CVD, ALD, PECVD, PVD, or the like. The adhesion layer **107** may alternately be referred to as a glue layer and improves adhesion between the conductive material **105** and the overlying layers of the gate electrode (e.g., the barrier layer **109** and first metal fill layer **117**).

(62) In FIGS. **20A** and **20B**, a barrier layer **109** is deposited conformally on the adhesion layer **107**. In some embodiments, the barrier layer **109** comprises tungsten and is deposited by an ALD process, or the like. In some embodiments, a process temperature of the ALD process may be in a range of about 250° C. to about 450 ° C. It has been observed that a deposition temperature less than about 250° C. may result in an unacceptably low deposition rate, negatively affecting manufacturing yield. It has also been observed that a deposition temperature greater than 450° C. may result in process tool damage (e.g., ALD deposition chamber damage). In some embodiments, the ALD process may be performed at a pressure of about 0.5 Torr to about 20 Torr. It has been observed that a deposition pressure outside of this range (e.g., less than about 0.5 Torr or greater than about 20 Torr) may result in an inability to precisely control the deposition process, negatively affecting yield.

(63) Further, the ALD process for the barrier layer **109** may include flowing one or more precursors into the deposition chamber, and the precursors used during the ALD process may be free of fluorine. For example, when the barrier layer **109** is a tungsten layer, the precursors used in the ALD process may include a combination of $W_{sub.x}Cl_{sub.y}$ and $H_{sub.2}$, where x and y are integers and $0 < x < 1$ and $0 < y < 1$. Because the ALD process does not use any fluorine-comprising precursors, the barrier layer **109** may be free of fluorine and prevents (or at least reduces) diffusion of fluorine from subsequently formed layers into the underlying adhesion layer **107** and the conductive material **105**. The barrier layer **109** may be deposited to a thickness $T_{sub.1}$ in a range of 15 Å to 90 Å. It has been observed that a thickness $T_{sub.1}$ less than 15 Å may not sufficiently prevent diffusion of fluorine, thereby negatively affecting threshold voltage stability and control in the completed transistor. It has also been observed that a thickness $T_{sub.1}$ greater than 90 Å, manufacturing costs and/or time may be unacceptably high as ALD processes are more costly and time consuming than other deposition processes (e.g., CVD).

(64) In FIGS. **21A**, **21B**, **21C**, **21D**, and **21E**, remaining portions of the gate electrodes **102** are deposited, and a fill metal **117** is formed to fill the remaining portions of the second recesses **98**. FIG. **21C** illustrates a detailed view of area **119** of FIG. **21B**. FIG. **21D** illustrates a top down view along line X-X' of FIG. **21B** (e.g., in the regions **501**), and FIG. **21E** illustrates a top down view along line Y-Y' of FIG. **21B** (e.g., through the first nanostructures **52**). The resulting gate electrodes **102** are formed for replacement gates and may comprise the conductive material **105**, the adhesion layer **107**, the barrier layer **109**, the first fill metal layer **117**, and the second fill metal layer **119**.

(65) The fill metal **117** may comprise cobalt, ruthenium, aluminum, tungsten, combinations thereof, or the like. In some embodiments, the fill metal **117** is deposited using two different deposition processes. For example, a first portion **117A** of fill metal layer **117** is deposited over the barrier layer **109** by an ALD process, and a second portion **117B** of the fill metal layer **117** may then be deposited over the first portion **117A** using a CVD process. It has been observed that the ALD process may allow for a more controlled deposition in areas around the nanostructures **52** while the CVD process provides a faster deposition rate for improved yield. In other embodiments, the fill metal **117** may be deposited using a single deposition process (e.g., CVD), and the first portion **117A** that is deposited by the ALD process may be omitted. Accordingly, the first portion **117A** is illustrated with dashed lines to indicate that it is optional.

(66) Depositing the fill metal **117** may include using a fluorine-comprising precursor in the ALD and/or CVD processes, and a material composition of the fill metal **117** may be uniform throughout the fill metal **117** regardless of whether a single deposition or multiple depositions are performed. For example, in embodiments where the fill metal **117** comprises tungsten, WF.sub.6 may be used as a precursor in the ALD and/or CVD processes. Because a fluorine comprising precursor is used to deposit the fill metal **117**, the fill metal **117** may also comprise fluorine. In various embodiments, a fluorine composition of the fill metal **117** may be higher than a fluorine composition of the barrier layer **109**. For example, the barrier layer **109** may be substantially free of fluorine, and a fluorine concentration of the fill metal **117** may be in a range of 3.5% to 25%. Thus, the fill metal **117** may be distinguished from the barrier layer **109** by a fluorine composition analysis, such as XPS depth profile analysis.

(67) Although the fill metal **117** comprises fluorine, the barrier layer **109** may prevent (or at least reduce) fluorine diffusion into the underlying adhesion layer **109** and conductive material **105**, and an uneven distribution of fluorine in the different regions (e.g., within the regions **501** compared to outside of the regions **501**) of the conductive material **105** can be avoided. As a result, threshold voltage control and stability in the resulting transistor can be improved by including the barrier layer **109**.

(68) In the p-type region **50P**, the gate dielectrics **100**, the conductive material **105** may be formed on top surfaces, sidewalls, and bottom surfaces of the first nanostructures **52**. The gate dielectrics **100**, the conductive material **105**, the adhesion layer **107**, the barrier layer **109**, and the fill metal **117** may also be deposited on top surfaces and sidewalls of the first nanostructures **52** as well as the top surfaces of the first ILD **96**, the CESL **94**, the first spacers **81**, and the STI regions **58**. After the filling of the second recesses **98**, a planarization process, such as a CMP, may be performed to remove the excess portions of the gate dielectrics **100**, the conductive material **105**, the adhesion layer **107**, the barrier layer **109**, and the fill metal **117** which excess portions are over the top surface of the first ILD **96**. The remaining portions of material of the gate electrodes **102** and the gate dielectrics **100** thus form replacement gate structures of the resulting nano-FETs. The gate electrodes **102** and the gate dielectrics **100** may be collectively referred to as “gate structures.”

(69) FIGS. 22A and 22B illustrate of a gate stack in the n-type region **50N**. Forming the gate stack in the n-type region **50N** may include first removing the first nanostructures **52** in the n-type region **50N**. The first nanostructures **52** may be removed by forming a mask (not shown) over the p-type region **50P** and performing an isotropic etching process such as wet etching or the like using etchants which are selective to the materials of the first nanostructures **52**, while the second nanostructures **54**, the substrate **50**, and the STI regions **68** remain relatively unetched as compared to the first nanostructures **52**. In embodiments in which the first nanostructures **52A-52C** include, e.g., SiGe, and the second nanostructures **54A-54C** include, e.g., Si or SiC, tetramethylammonium hydroxide (TMAH), ammonium hydroxide (NH.sub.4OH) or the like may be used to remove the first nanostructures **52** in the n-type region **50N**.

(70) The gate stack is then formed over and around the second nanostructures **54** in the n-type region **50N**. The gate stack includes the gate dielectrics **100** and gate electrodes **127**. In some

embodiments, the gate dielectrics **100** in the n-type region **50N** and the p-type region **50P** may be formed simultaneously. Further, at least portions of the gate electrodes **127** may be formed either before or after forming the gate electrodes **102** (see FIGS. **20A** and **20B**), and at least portions of the gate electrodes **127** may be formed while the p-type region **50P** is masked. As such, the gate electrodes **127** may comprise different materials than the gate electrodes **102**. For example, the gate electrodes **127** may comprise a conductive layer **121**, a barrier layer **123**, and a fill metal **125**. The conductive layer **121** may be an n-type work function metal (WFM) layer comprising an n-type metal, such as, titanium aluminum, titanium aluminum carbide, tantalum aluminum, tantalum carbide, combinations thereof, or the like. The conductive layer **121** may be deposited by CVD, ALD, PECVD, PVD, or the like. The barrier layer **123** may comprise titanium nitride, tantalum nitride, tungsten carbide, combinations thereof, or the like, and the barrier layer **123** may further function as an adhesion layer. The barrier layer **123** may be deposited by CVD, ALD, PECVD, PVD, or the like. The fill metal **125** comprises cobalt, ruthenium, aluminum, tungsten, combinations thereof, or the like, which is deposited by CVD, ALD, PECVD, PVD, combinations thereof, or the like. The fill metal **125** may or may not have a same material composition and be deposited concurrently with the fill metal **117**. Although not explicitly illustrated, an additional barrier layer (e.g., similar to the barrier layer **109** discussed above) may be deposited between the barrier layer **123** and the fill metal **125** to prevent fluorine diffusion from the fill metal **125** into the underlying barrier layer **123** and/or the conductive layer **121**.

(71) After the filling of the second recesses **98**, a planarization process, such as a CMP, may be performed to remove the excess portions of the gate dielectrics **100** and the gate electrodes **127**, which excess portions are over the top surface of the first ILD **96**. The remaining portions of material of the gate electrodes **127** and the gate dielectrics **100** thus form replacement gate structures of the resulting nano-FETs of the n-type region **50N**. The CMP processes to remove excess materials of the gate electrodes **102** in the p-type region **50P** and to remove excess materials of the gate electrodes **127** in the n-type region **50N** may be performed concurrently or separately.

(72) In FIGS. **23A-23C**, the gate structure (including the gate dielectrics **100**, the gate electrodes **102**, and the gate electrodes **127**) is recessed, so that a recess is formed directly over the gate structure and between opposing portions of first spacers **81**. A gate mask **104** comprising one or more layers of dielectric material, such as silicon nitride, silicon oxynitride, or the like, is filled in the recess, followed by a planarization process to remove excess portions of the dielectric material extending over the first ILD **96**. Subsequently formed gate contacts (such as the gate contacts **114**, discussed below with respect to FIGS. **24A** and **24B**) penetrate through the gate mask **104** to contact the top surface of the recessed gate electrodes **102**.

(73) As further illustrated by FIGS. **23A-23C**, a second ILD **106** is deposited over the first ILD **96** and over the gate mask **104**. In some embodiments, the second ILD **106** is a flowable film formed by FCVD. In some embodiments, the second ILD **106** is formed of a dielectric material such as PSG, BSG, BPSG, USG, or the like, and may be deposited by any suitable method, such as CVD, PECVD, or the like.

(74) In FIGS. **24A-24C**, the second ILD **106**, the first ILD **96**, the CESL **94**, and the gate masks **104** are etched to form third recesses **108** exposing surfaces of the epitaxial source/drain regions **92** and/or the gate structure. The third recesses **108** may be formed by etching using an anisotropic etching process, such as RIE, NBE, or the like. In some embodiments, the third recesses **108** may be etched through the second ILD **106** and the first ILD **96** using a first etching process; may be etched through the gate masks **104** using a second etching process; and may then be etched through the CESL **94** using a third etching process. A mask, such as a photoresist, may be formed and patterned over the second ILD **106** to mask portions of the second ILD **106** from the first etching process and the second etching process. In some embodiments, the etching process may over-etch, and therefore, the third recesses **108** extend into the epitaxial source/drain regions **92** and/or the gate structure, and a bottom of the third recesses **108** may be level with (e.g., at a same level, or

having a same distance from the substrate), or lower than (e.g., closer to the substrate) the epitaxial source/drain regions **92** and/or the gate structure. Although FIG. **24B** illustrates the third recesses **108** as exposing the epitaxial source/drain regions **92** and the gate structure in a same cross section, in various embodiments, the epitaxial source/drain regions **92** and the gate structure may be exposed in different cross-sections, thereby reducing the risk of shorting subsequently formed contacts.

(75) After the third recesses **108** are formed, silicide regions **110** are formed over the epitaxial source/drain regions **92**. In some embodiments, the silicide regions **110** are formed by first depositing a metal (not shown) capable of reacting with the semiconductor materials of the underlying epitaxial source/drain regions **92** (e.g., silicon, silicon germanium, germanium) to form silicide or germanide regions, such as nickel, cobalt, titanium, tantalum, platinum, tungsten, other noble metals, other refractory metals, rare earth metals or their alloys, over the exposed portions of the epitaxial source/drain regions **92**, then performing a thermal anneal process to form the silicide regions **110**. The un-reacted portions of the deposited metal are then removed, e.g., by an etching process. Although silicide regions **110** are referred to as silicide regions, silicide regions **110** may also be germanide regions, or silicon germanide regions (e.g., regions comprising silicide and germanide). In an embodiment, the silicide region **110** comprises TiSi, and has a thickness in a range between about 2 nm and about 10 nm.

(76) Next, in FIGS. **25A-C**, contacts **112** and **114** (may also be referred to as contact plugs) are formed in the third recesses **108**. The contacts **112** and **114** may each comprise one or more layers, such as barrier layers, diffusion layers, and fill materials. For example, in some embodiments, the contacts **112** and **114** each include a barrier layer and a conductive material, and is electrically coupled to the underlying conductive feature (e.g., the gate electrodes **102**, the gate electrodes **127**, and/or silicide region **110** in the illustrated embodiment). The contacts **114** are electrically coupled to the gate electrodes **102** and **127** and may be referred to as gate contacts, and the contacts **112** are electrically coupled to the silicide regions **110** and may be referred to as source/drain contacts. The barrier layer may include titanium, titanium nitride, tantalum, tantalum nitride, or the like. The conductive material may be copper, a copper alloy, silver, gold, tungsten, cobalt, aluminum, nickel, or the like. A planarization process, such as a CMP, may be performed to remove excess material from a surface of the second ILD **106**.

(77) FIGS. **26A** through **26C** illustrate cross-sectional views of a device according to some alternative embodiments. FIG. **26A** illustrates reference cross-section A-A' illustrated in FIG. **1**. FIG. **26B** illustrates reference cross-section B-B' illustrated in FIG. **1**. FIG. **26C** illustrates reference cross-section C-C' illustrated in FIG. **1**. In FIGS. **26A-C**, like reference numerals indicate like elements formed by like processes as the structure of FIGS. **25A-C**. However, in FIGS. **26A-C**, channel regions in the n-type region **50N** and the p-type region **50P** comprise a same material. For example, the second nanostructures **54**, which comprise silicon, provide channel regions for p-type NSFETs in the p-type region **50P** and for n-type NSFETs in the n-type region **50N**. The structure of FIGS. **26A-C** may be formed, for example, by removing the first nanostructures **52** from both the p-type region **50P** and the n-type region **50N** simultaneously; depositing the gate dielectrics **100** and the gate electrodes **102** around the second nanostructures **54** in the p-type region **50P**; and depositing the gate dielectrics **100** and the gate electrodes **104** around the first nanostructures **54** in the n-type region **50N**.

(78) FIGS. **27A** through **27C** illustrate cross-sectional views of a device according to some alternative embodiments. FIG. **27A** illustrates reference cross-section A-A' illustrated in FIG. **1**. FIG. **27B** illustrates reference cross-section B-B' illustrated in FIG. **1**. FIG. **27C** illustrates reference cross-section C-C' illustrated in FIG. **1**. In FIGS. **27A-C**, like reference numerals indicate like elements formed by like processes as the structure of FIGS. **25A-C**. However, in FIGS. **27A-C**, the fill metal **117** is omitted, and the material of the barrier layer **109** is deposited to fill the second recesses **98** and provide the fill material of the gate electrodes. As a result, the gate electrodes are

formed to be substantially free of any fluorine. For example, fluorine-free tungsten may be used as a fill metal over the adhesion layer **107**.

(79) Various embodiments provide transistor gate stacks where a conductive material (e.g., tungsten) is deposited as a fill metal. Various embodiments may further include a barrier layer between the fill metal and underlying work function metal (WFM) layers to prevent fluorine diffusion during deposition and/or the fill metal into the underlying WFM layers. For example, the barrier layer may comprise a tungsten layer that is deposited with fluorine free precursors. By preventing fluorine diffusion into the underlying layers, advantages may be achieved. For example, by preventing fluorine diffusion into the underlying layers, uneven fluorine distribution in the WFM layers may be avoided in different areas of the gate stack, which results in improved threshold voltage (V_t) stability and control. V_t stability and control may be especially improved in nano-FETs, which are susceptible to uneven fluorine diffusion in regions of the WFM layers between nanosheets compared to other regions of the WFM layers.

(80) In some embodiments, a device includes a first nanostructure; a second nanostructure over the first nanostructure; a high-k gate dielectric around the first nanostructure and the second nanostructure, the high-k gate dielectric having a first portion on a top surface of the first nanostructure and a second portion on a bottom surface of the second nanostructure; and a gate electrode over the high-k gate dielectric. The gate electrode comprises: a first work function metal around the first nanostructure and the second nanostructure, the first work function metal filling a region between the first portion of the high-k gate dielectric and the second portion of the high-k gate dielectric; and a tungsten layer over the first work function metal, the tungsten layer being free of fluorine. Optionally, in some embodiments, the gate electrode further comprises a fill material over the tungsten layer, the fill material having a higher concentration of fluorine than the tungsten layer. Optionally, in some embodiments, the fill material comprises tungsten. Optionally, in some embodiments, a thickness of tungsten layer is in a range of 15 Å to 90 Å. Optionally, in some embodiments, the first work function metal is free of fluorine. Optionally, in some embodiments, the gate electrode further comprises an adhesion layer between the tungsten layer and the first work function metal.

(81) In some embodiments, a transistor includes a first nanostructure over a semiconductor substrate; a second nanostructure over the first nanostructure; a gate dielectric surrounding the first nanostructure and the second nanostructure; and a gate electrode over the gate dielectric. The gate electrode comprises: a work function metal around the first nanostructure and the second nanostructure; a barrier layer on the work function metal; and a fill metal over the barrier layer, wherein the fill metal has a higher concentration of fluorine than the barrier layer. Optionally, in some embodiments, the transistor further comprises an adhesion layer between the barrier layer and the fill metal. Optionally, in some embodiments, a thickness of the barrier layer is in a range of 15 Å to 90 Å. Optionally, in some embodiments, the barrier layer is free of fluorine. Optionally, in some embodiments, the work function metal is free of fluorine. Optionally, in some embodiments, the barrier layer and the fill metal each comprise tungsten. Optionally, in some embodiments, the transistor further comprises an interfacial layer under the gate dielectric, the interface layer surrounding the first nanostructure and the second nanostructure, and the gate dielectric comprises a high-k material.

(82) In some embodiments, a method includes depositing a gate dielectric around a first nanostructure and a second nanostructure, the first nanostructure is disposed over the second nanostructure; depositing a work function metal over the gate dielectric; depositing a barrier layer over the work function metal, wherein depositing the barrier layer comprises a first atomic layer deposition (ALD) process, and wherein the first ALD process does not use any fluorine-comprising precursors; and depositing a fill metal over the barrier layer, wherein depositing the fill metal comprises using a fluorine-comprising precursor. Optionally, in some embodiments, the first ALD process comprises using a combination of $W_{\text{sub}}.xCl_{\text{sub}}.y$ and $H_{\text{sub}}.2$ as precursors. Optionally,

in some embodiments, depositing the fill metal comprises a chemical vapor deposition (CVD) process, and wherein the fluorine-comprising precursor is WF.sub.6. Optionally, in some embodiments, depositing the fill metal further comprises a second ALD process performed before the CVD process. Optionally, in some embodiments, the ALD process comprises using WF.sub.6 as a precursor. Optionally, in some embodiments, the method further includes depositing an adhesion layer between the work function metal and the barrier layer. Optionally, in some embodiments, the barrier layer and the fill metal each comprise tungsten.

(83) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A device comprising: a first semiconductor structure; a second semiconductor structure over the first semiconductor structure; a high-k gate dielectric around the first semiconductor structure and the second semiconductor structure; and a gate electrode over the high-k gate dielectric, wherein the gate electrode comprises: a first work function metal around the first semiconductor structure and the second semiconductor structure, the first work function metal comprising an internal seam disposed between the first semiconductor structure and the second semiconductor structure, wherein a width of the internal seam in a first cross-sectional view is less than a width of the first semiconductor structure in the first cross-sectional view; and a barrier material over the first work function metal, wherein the barrier material comprises tungsten.
2. The device of claim 1, wherein the gate electrode further comprises a fill material over the barrier material, the fill material having a higher concentration of fluorine than the barrier material.
3. The device of claim 2, wherein the fill material comprises tungsten.
4. The device of claim 2, wherein the fill material has a higher concentration of fluorine than the first work function metal.
5. The device of claim 1, wherein the first work function metal is a p-type work function metal.
6. The device of claim 1, wherein the gate electrode further comprises an adhesion layer between the barrier material and the first work function metal.
7. The device of claim 1 further comprising a second work function metal between the first work function metal and the high-k gate dielectric, wherein the second work function metal has a same conductivity type as the first work function metal.
8. A transistor comprising: a first nanostructure over a semiconductor substrate; a second nanostructure over the first nanostructure; a gate dielectric surrounding the first nanostructure and the second nanostructure, wherein a first portion of the gate dielectric on a bottom surface of the first nanostructure is spaced apart from a second portion of the gate dielectric on a top surface of the second nanostructure; and a gate electrode over the gate dielectric, wherein the gate electrode comprises: a work function metal around the first nanostructure and the second nanostructure, wherein the work function metal spans an entire distance from the first portion of the gate dielectric to the second portion of the gate dielectric; a barrier metal over the work function metal, the barrier metal comprising tungsten; and a fill metal over the barrier metal.
9. The transistor of claim 8, wherein a thickness of the barrier metal is in a range of 15 Å to 90 Å.
10. The transistor of claim 8, wherein the barrier metal has a lower concentration of fluorine than the fill metal.

11. The transistor of claim 8, wherein the work function metal is a p-type work function metal, and wherein a first portion of the work function metal meets a second portion of the work function metal at a seam between the first nanostructure and the second nanostructure.
 12. The transistor of claim 8, wherein the fill metal comprises tungsten.
 13. The transistor of claim 8, wherein the work function metal comprises: a first work function metal; and a second work function metal on the first work function metal, wherein the first work function metal and the second work function metal are made of different materials, and wherein the first work function metal and the second work function metal have a same conductivity type.
 14. A method comprising: depositing a gate dielectric around a first nanostructure and a second nanostructure, the first nanostructure is disposed over the second nanostructure; depositing a work function metal over the gate dielectric, wherein the work function metal completely fills a region extending from a first portion of the gate dielectric on a bottom surface of the first nanostructure to a second portion of the gate dielectric on a top surface of the second nanostructure; depositing a barrier layer over the work function metal, wherein the barrier layer comprises tungsten; and depositing a fill metal over the barrier layer.
 15. The method of claim 14, wherein depositing the barrier layer comprises a fluorine-free deposition process.
 16. The method of claim 15, wherein depositing the fill metal comprises flowing a fluorine-comprising precursor.
 17. The method of claim 16, wherein the barrier layer reduces fluorine diffusion into the work function metal while depositing the fill metal.
 18. The method of claim 14, wherein depositing the fill metal comprises: performing a first deposition process to deposit a first portion of the fill metal; and performing a second deposition process to deposit a second portion of the fill metal over the first portion of the fill metal, wherein the second deposition process is a different type of deposition process than the first deposition process.
 19. The method of claim 18, wherein the first deposition process is an atomic layer deposition (ALD) process, and wherein the second deposition process is a chemical vapor deposition (CVD) process.
 20. The method of claim 14, wherein depositing the barrier layer is performed at a temperature in a range of 250° C. to 450° C.
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